
SEMIRING TENSOR NETWORKS

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

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1 Semiring

Definition 1. A semiring is a tuple $(S, \oplus, \otimes, 0, 1)$, where S is a set, $0, 1 \in S$, and

$$\oplus : S \times S \rightarrow S \quad , \quad \otimes : S \times S \rightarrow S ,$$

such that the following properties hold:

(i) $(S, \oplus, 0)$ is a commutative monoid.

(ii) $(S, \otimes, 1)$ is a commutative monoid.

(iii) $\otimes$ distributes over $\oplus$, that is for any $s_0, s_1, s_2 \in S$

$$s_0 \otimes (s_1 \oplus s_2) = (s_0 \otimes s_1) \oplus (s_0 \otimes s_2) .$$

(iv) 0 annihilates $\otimes$, that is (check!)

$$0 \otimes (s_0 \otimes s_1) = 0 .$$

We understand S as the set of coordinates in a tensor.

Definition 2. A tensor with coordinates in S of order $d \in \mathbb{N}$ and leg dimensions $m_{[d]} = (m_0, \dots, m_{d-1})$, where $m_k \in \mathbb{N}$ for $k \in [d]$, is a map

$$\tau : \prod_{k \in [d]} [m_k] \rightarrow S .$$

A tensor network with coordinates in S is a hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where each node is decorated by a dimension m_v and each edge by a tensor of.

2 Contractions

Definition 3. Given a semiring $(S, \oplus, \otimes, 0, 1)$ and a tensor network $\{\tau^e [X_e] : e \in \mathcal{E}\}$ with coordinates in S , the contraction of the tensor network keeping the variables to a subset $\mathcal{U} \subset \mathcal{V}$ open is the tensor

$$\langle \{\tau^e [X_e] : e \in \mathcal{E}\} \rangle_{X_{\mathcal{U}}}^{(S, \oplus, \otimes, 0, 1)}$$

with coordinates at indices $x_{\mathcal{U}}$ by

$$\langle \{\tau^e [X_e] : e \in \mathcal{E}\} \rangle_{X_{\mathcal{U}}}^{(S, \oplus, \otimes, 0, 1)} = \bigoplus_{x_{\mathcal{V} \setminus \mathcal{U}}} \bigotimes_{e \in \mathcal{E}} \tau^e [X_e = x_e] .$$

The distributive property (iii) of the semiring ensures, that we can compute contractions by sub-contractions.

Theorem 1. Given a disjoint partition of $\mathcal{E}$ into $\mathcal{E}^0, \mathcal{E}^1$, and a subset $\mathcal{U} \subset \mathcal{V}$, we define the sets $\mathcal{V}^0, \mathcal{V}^1$ and $\mathcal{V}^{0 \cup 1}$ as the nodes appearing in $\mathcal{E}^0$, respectively $\mathcal{E}^1$ and in $\mathcal{E}^0$ and $\mathcal{E}^1$. We then have

$$\langle \{\tau^e [X_e] : e \in \mathcal{E}\} \rangle_{X_{\mathcal{U}}}^{(S, \oplus, \otimes, 0, 1)} = \langle \langle \{\tau^e [X_e] : e \in \mathcal{E}^0\} \rangle_{X_{\mathcal{V}^0 \cap \mathcal{U}}, X_{\mathcal{V}^{0 \cup 1}}}^{(S, \oplus, \otimes, 0, 1)}, \langle \{\tau^e [X_e] : e \in \mathcal{E}^1\} \rangle_{X_{\mathcal{V}^1 \cap \mathcal{U}}, X_{\mathcal{V}^{0 \cup 1}}}^{(S, \oplus, \otimes, 0, 1)} \rangle_{X_{\mathcal{U}}}^{(S, \oplus, \otimes, 0, 1)} .$$

Proof. We use the commutative properties to split the summations and the distributive property to move the $\bigoplus$ to closed variables only in $\mathcal{V}^0$ respectively $\mathcal{V}^1$ to the first respectively second sub-contraction. $\square$

3 Examples

3.1 Standard TN

Standard TN use the semiring

$$(\mathbb{R}, +, \cdot, 0, 1) .$$

3.2 Constraint Satisfaction

We take $S = \{0, 1\}$ and interpret 0 as False, 1 as True. We define the disjunction and conjunction operations by

$$\vee [X_0, X_1] = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} [X_0, X_1]$$

and

$$\wedge [X_0, X_1] = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} [X_0, X_1] .$$

Tensors over $\{0, 1\}$ are interpreted as constraints, and their tensor networks as constraint networks.

The satisfiability of a constraint network $\{\tau^e [X_e] : e \in \mathcal{E}\}$ is the contraction

$$\bigvee_{x_V} \bigwedge_{e \in \mathcal{E}} \tau^e [X_e = x_e] = \langle \{\tau^e [X_e] : e \in \mathcal{E}\} \rangle_{\emptyset}^{\langle \{0,1\}, \vee, \wedge, 0, 1 \rangle} .$$

3.3 Maximum calculus

The maximal coordinate of a standard TN is the contraction wrt the semiring

$$(\mathbb{R}, \max, \cdot, 0, 1) .$$

Analogously the minimal coordinate by

$$(\mathbb{R}, \min, \cdot, 0, 1) .$$

These schemes only work, when taking the minimal coordiante keeping all variables open. When searching for the maximal/minimal coordiante of a contraction with some variables closed, we need to first coarse grain the tensor network.

These contractions are used in encoding-decoding schemes Mézard (2009); MacKay (2003).

References

- David J. C. MacKay. *Information Theory, Inference and Learning Algorithms*. Cambridge University Press, Cambridge, illustrated edition edition, September 2003. ISBN 978-0-521-64298-9.
- Marc Mézard. *Information, Physics, and Computation*. ACADEMIC, Oxford ; New York, March 2009. ISBN 978-0-19-857083-7.